

Mathematical Customer Lifetime Value (CLV) Projection via BG/NBD Model from Scratch

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Abstract

This project presents the implementation of the BG/NBD (Beta Geometric/Negative Binomial Distribution) model from first principles using Python. The model estimates customer purchasing behavior, predicts future transactions, and projects Customer Lifetime Value (CLV). The implementation uses NumPy for numerical computation and SciPy's L-BFGS-B optimization algorithm for parameter estimation without relying on specialized marketing libraries.

Introduction

Customer Lifetime Value (CLV) is an important business metric used to estimate the total value a customer generates throughout their relationship with a company. Organizations use CLV to understand customer behavior, improve customer retention, and increase business profitability. The BG/NBD (Beta Geometric/Negative Binomial Distribution) model is a widely used statistical model for predicting future customer transactions. In this project, the BG/NBD model is implemented from scratch using Python, NumPy, and SciPy without relying on specialized marketing libraries.

Theory

The BG/NBD model assumes that customer purchase behavior follows a Poisson process with Gamma-distributed transaction rates. Customer dropout probability follows a Beta distribution. The model captures customer heterogeneity and estimates the

probability that a customer is still active. Based on these assumptions, it predicts future transactions and helps estimate Customer Lifetime Value (CLV).

Objectives

- Implement the BG/NBD model from scratch.
- Estimate the parameters r , α , a , and b using Maximum Likelihood Estimation (MLE).
- Predict future customer transactions.
- Calculate Customer Lifetime Value (CLV).
- Understand customer purchasing behavior using statistical modeling.

Methodology

The implementation follows a structured approach to estimate Customer Lifetime Value using the BG/NBD model.

1. Collect and prepare customer transaction data.
2. Generate customer summary variables (x , tx , T).

- 3. Compute the vectorized negative log-likelihood function.**
- 4. Estimate model parameters using SciPy's L-BFGS-B optimization algorithm.**
- 5. Predict future customer transactions.**
- 6. Calculate Customer Lifetime Value (CLV).**

Dataset

A synthetic transaction dataset is used for testing and validating the BG/NBD implementation. The same approach can also be applied to real customer transaction data after converting it into customer summary variables such as purchase frequency (x), recency (tx), and observation period (T).

Likelihood

The BG/NBD model estimates customer purchasing behavior by maximizing the log-likelihood function. Maximum Likelihood Estimation (MLE) is used to estimate the model parameters (r , α , a , and b). The optimization process minimizes the negative log-likelihood using SciPy's L-BFGS-B algorithm.

Complexity Analysis

The implementation uses vectorized NumPy operations to improve computational efficiency. The likelihood function is evaluated in $O(N)$ time for each optimization iteration, making the implementation suitable for large transaction datasets while avoiding slow Python loops.

Results

The BG/NBD implementation successfully estimates the model parameters (r , α , a , and b). Based on these estimated parameters, the model predicts the probability that a customer is active, estimates the expected number of future transactions, and projects Customer Lifetime Value (CLV). The implementation demonstrates efficient performance using vectorized numerical computations.

Conclusion

This project successfully demonstrates the implementation of the BG/NBD model from first principles using Python, NumPy, and SciPy. The model provides an effective approach for predicting future customer transactions and estimating Customer Lifetime Value. The implementation is numerically efficient, scalable, and suitable for educational and research purposes.

References

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